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Midterm – Machine Learning

Presentation of midterm solution

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Introduction

This report compares supervised classification and unsupervised clustering on the Breast Cancer Wisconsin and Wine datasets. Both datasets were processed with the same fixed experimental plan: three training-validation seeds, one fixed test set, unscaled and StandardScaler representations, Decision Tree, KNN, Gaussian Naive Bayes, and K-Means models. Model selection used validation results only, while final test evaluation used the predeclared seed-42 fitted models.

Loading the Datasets

The Breast Cancer Wisconsin dataset contains 569 observations, 30 numerical input features and two classes. Class 0 represents malignant cases and class 1 represents benign cases. The Wine dataset contains 178 observations, 13 numerical input features and three original class labels: 0, 1 and 2. In both datasets, the target column was separated from the feature matrix before analysis.

Dataset Quality Checks

No missing values were detected in either dataset, so imputation was not applied. Duplicate-row counts and feature data types were inspected before modelling. All input features were numerical. Potential outliers were identified with the IQR rule and retained because automatic deletion or clipping could remove meaningful observations and change model behaviour. No missing values and no duplicate rows were found in either dataset. All input features were numerical, so no categorical encoding was required. Potential outliers were retained because deleting or clipping them could remove meaningful observations and change model behaviour.

Exploratory Data Analysis

Descriptive statistics were calculated for every feature. The visual analysis below compares class balance, Pearson feature correlations and feature ranges. The boxplots show that several variables have very different scales, supporting the use of a scaled representation alongside the original data. For the Breast Cancer dataset, the three strongest unique feature correlations were mean radius–mean perimeter (0.9979), worst radius–worst perimeter (0.9937), and mean radius–mean area (0.9874). These high correlations show that several measurements contain similar size-related information.

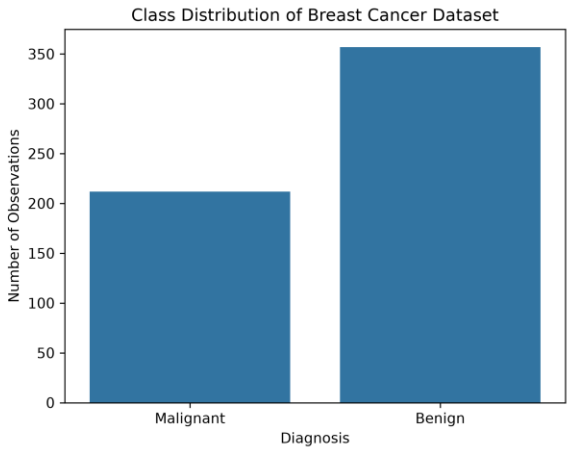


Figure 1 Breast Cancer class distribution

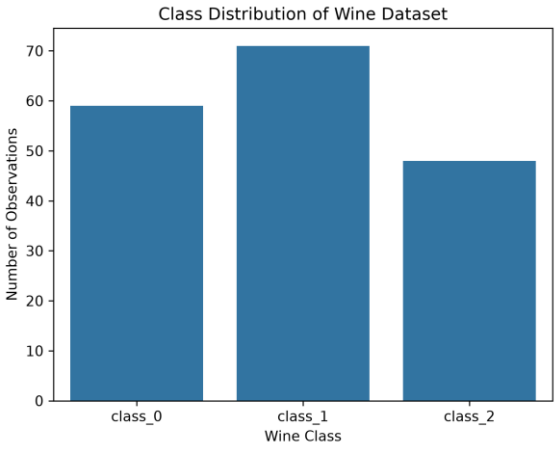


Figure 2 Wine class distribution

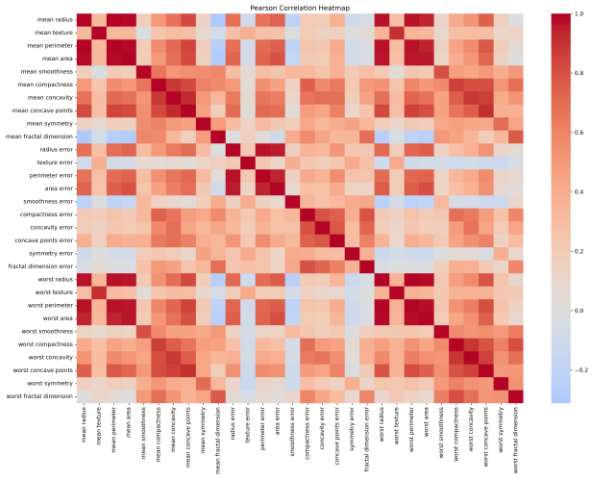


Figure 3 Breast Cancer Pearson correlation heatmap

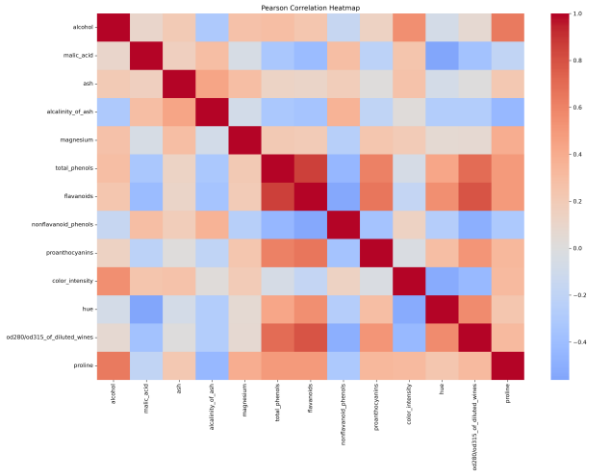


Figure 4 Wine Pearson correlation heatmap

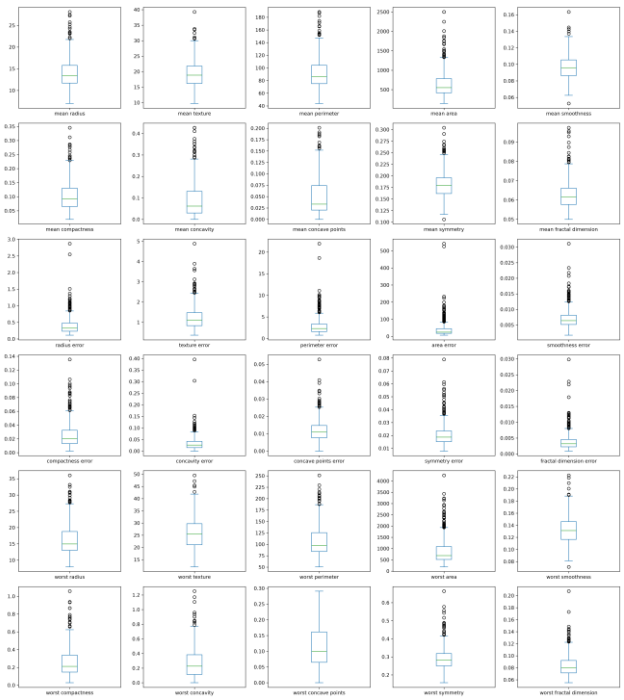


Figure 5 Breast Cancer feature boxplots

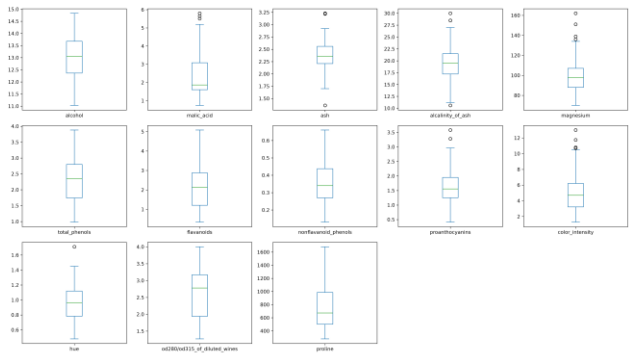


Figure 6 Wine feature boxplots

The boxplots show that area- and perimeter-related variables have much larger numerical ranges than ratio- and error-related variables. Several features also contain potential outliers. This scale difference is especially important for distance-based algorithms such as KNN and K-Means.

Splitting the Datasets

For each dataset, 20% of the observations were reserved once as a fixed stratified test set using random_state=42. The remaining 80% was divided into approximately 70% training and 10% validation subsets under seeds 42, 52 and 62. Original row indices were preserved, no subset overlap occurred within a repetition, and the same test indices were retained for all three repetitions.

Data Preprocessing and Scaling

StandardScaler was selected for both datasets. A separate scaler was fitted for each dataset and seed using only the corresponding training subset. Validation data were transformed with the matching training-fitted scaler. The final test representation used only the scaler fitted under seed 42. This procedure prevented validation and test information from influencing preprocessing. For seed 42, the before-and-after displays confirmed that the original feature values were transformed to approximately zero-centred and unit-scale values. The row indices and feature order remained unchanged after scaling.

Preparing the Two Data Representations

Every experiment used two representations with identical rows and columns: the original unscaled features and the StandardScaler-transformed features. No feature selection or dimensionality reduction was applied.

Training the Classification Models

Each dataset included 30 Decision Tree fits, 30 KNN fits and 6 Gaussian Naive Bayes fits. Therefore, 66 classifiers were fitted per dataset and 132 classifier models were fitted overall. Training accuracy and training macro-F1 were recorded for every run.

Measuring Validation Performance

Validation accuracy, macro-precision, macro-recall, macro-F1 and ROC-AUC were calculated for all classifier runs. Configurations were ranked by mean validation macro-F1, followed by mean validation accuracy, lower macro-F1 standard deviation and configuration ID. The required parameter-comparison figures show the mean training and validation macro-F1 values with standard-deviation error bars.

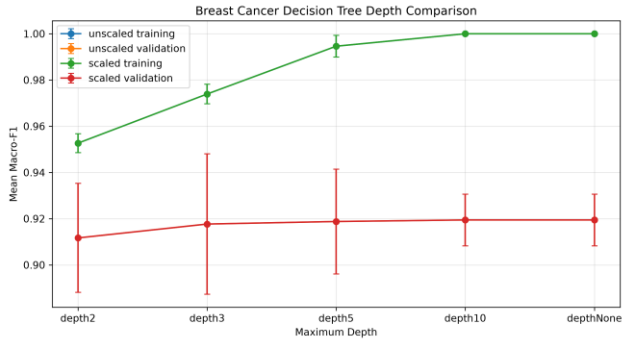


Figure 7 Breast Cancer Decision Tree depth comparison

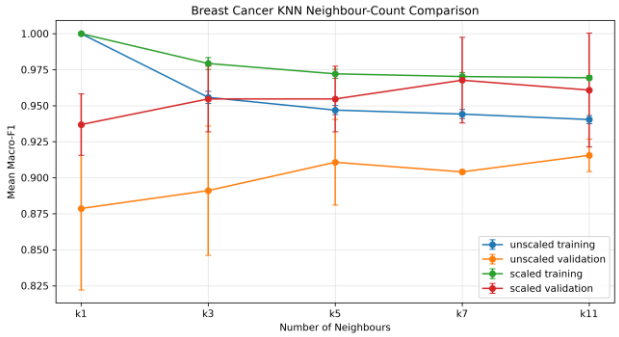


Figure 8 Breast Cancer KNN neighbour comparison

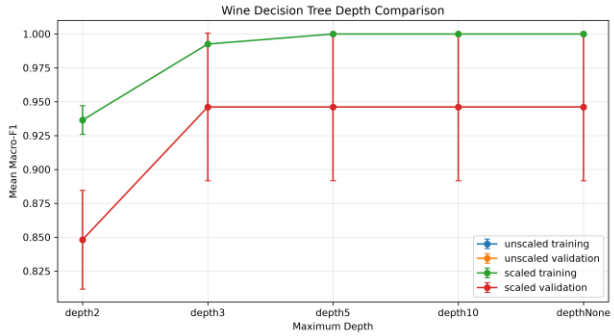


Figure 9 Wine Decision Tree depth comparison

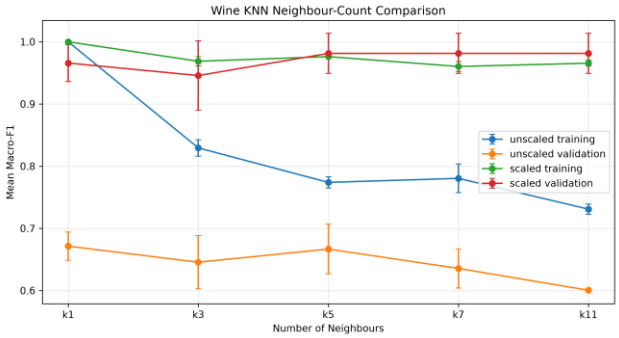


Figure 10 Wine KNN neighbour comparison

Classifier Summary Tables

Breast Cancer classifier summary

Algorithm	Representation	Parameter	Train Macro-F1 mean	Train Macro-F1 std	Val Accuracy mean	Val Macro-F1 mean	Val Macro-F1 std	Val ROC-AUC mean
dt	scaled	depth10	1.0000	0.0000	0.9240	0.9195	0.0112	0.9266
dt	scaled	depth2	0.9527	0.0041	0.9181	0.9117	0.0236	0.9220
dt	scaled	depth3	0.9739	0.0042	0.9240	0.9177	0.0304	0.9414
dt	scaled	depth5	0.9946	0.0047	0.9240	0.9188	0.0227	0.9301
dt	scaled	depthNone	1.0000	0.0000	0.9240	0.9195	0.0112	0.9266
dt	unscaled	depth10	1.0000	0.0000	0.9240	0.9195	0.0112	0.9266
dt	unscaled	depth2	0.9527	0.0041	0.9181	0.9117	0.0236	0.9220
dt	unscaled	depth3	0.9739	0.0042	0.9240	0.9177	0.0304	0.9414
dt	unscaled	depth5	0.9946	0.0047	0.9240	0.9188	0.0227	0.9301
dt	unscaled	depthNone	1.0000	0.0000	0.9240	0.9195	0.0112	0.9266
gnb	scaled	default	0.9307	0.0058	0.9298	0.9248	0.0201	0.9760
gnb	unscaled	default	0.9323	0.0049	0.9415	0.9369	0.0226	0.9797
knn	scaled	k1	1.0000	0.0000	0.9415	0.9369	0.0213	0.9339
knn	scaled	k11	0.9694	0.0015	0.9649	0.9608	0.0395	0.9945
knn	scaled	k3	0.9793	0.0041	0.9591	0.9547	0.0228	0.9793
knn	scaled	k5	0.9721	0.0032	0.9591	0.9547	0.0228	0.9795
knn	scaled	k7	0.9702	0.0028	0.9708	0.9677	0.0297	0.9782
knn	unscaled	k1	1.0000	0.0000	0.8889	0.8786	0.0566	0.8757
knn	unscaled	k11	0.9404	0.0028	0.9240	0.9155	0.0113	0.9605
knn	unscaled	k3	0.9558	0.0042	0.9006	0.8910	0.0449	0.9299
knn	unscaled	k5	0.9469	0.0032	0.9181	0.9107	0.0297	0.9480
knn	unscaled	k7	0.9442	0.0032	0.9123	0.9040	0.0013	0.9581

Wine classifier summary

Algorithm	Representation	Parameter	Train Macro-F1 mean	Train Macro-F1 std	Val Accuracy mean	Val Macro-F1 mean	Val Macro-F1 std	Val ROC-AUC mean
dt	scaled	depth10	1.0000	0.0000	0.9444	0.9462	0.0545	0.9552
dt	scaled	depth2	0.9365	0.0106	0.8519	0.8481	0.0365	0.9050
dt	scaled	depth3	0.9926	0.0000	0.9444	0.9462	0.0545	0.9611
dt	scaled	depth5	1.0000	0.0000	0.9444	0.9462	0.0545	0.9552
dt	scaled	depthNone	1.0000	0.0000	0.9444	0.9462	0.0545	0.9552
dt	unscaled	depth10	1.0000	0.0000	0.9444	0.9462	0.0545	0.9552
dt	unscaled	depth2	0.9365	0.0106	0.8519	0.8481	0.0365	0.9050
dt	unscaled	depth3	0.9926	0.0000	0.9444	0.9462	0.0545	0.9611
dt	unscaled	depth5	1.0000	0.0000	0.9444	0.9462	0.0545	0.9552
dt	unscaled	depthNone	1.0000	0.0000	0.9444	0.9462	0.0545	0.9552
gnb	scaled	default	0.9840	0.0005	0.9630	0.9638	0.0314	1.0000
gnb	unscaled	default	0.9840	0.0005	0.9630	0.9638	0.0314	1.0000
knn	scaled	k1	1.0000	0.0000	0.9630	0.9658	0.0296	0.9749
knn	scaled	k11	0.9656	0.0048	0.9815	0.9814	0.0323	1.0000

knn	scaled	k3	0.9687	0.0075	0.9444	0.9457	0.0558	1.0000
knn	scaled	k5	0.9762	0.0005	0.9815	0.9814	0.0323	1.0000
knn	scaled	k7	0.9603	0.0080	0.9815	0.9814	0.0323	1.0000
knn	unscaled	k1	1.0000	0.0000	0.7037	0.6714	0.0226	0.7654
knn	unscaled	k11	0.7309	0.0083	0.6296	0.6005	0.0010	0.8216
knn	unscaled	k3	0.8294	0.0134	0.6667	0.6456	0.0429	0.8333
knn	unscaled	k5	0.7739	0.0091	0.7037	0.6667	0.0399	0.8645
knn	unscaled	k7	0.7804	0.0231	0.6667	0.6356	0.0316	0.8459

Evaluating the Selected Models on the Test Sets

Breast Cancer selected knn_scaled_k7 and achieved test accuracy 0.9737, macro-F1 0.9713 and ROC-AUC 0.9888. Wine selected knn_scaled_k11 and achieved 1.0000 for test accuracy, macro-F1 and ROC-AUC. For Wine, class 0 was selected for the manual one-versus-rest check. The test confusion matrix contained no classification errors. Therefore, precision, recall and F1-score for class 0 were all 1.0000. These values matched the corresponding class-0 values in the classification report.

Dataset	Selected Configuration	Seed	Test Accuracy	Test Macro-Precision	Test Macro-Recall	Test Macro-F1	Test ROC-AUC
cancer	knn_scaled_k7	42	0.9737	0.9800	0.9643	0.9713	0.9888
wine	knn_scaled_k11	42	1.0000	1.0000	1.0000	1.0000	1.0000

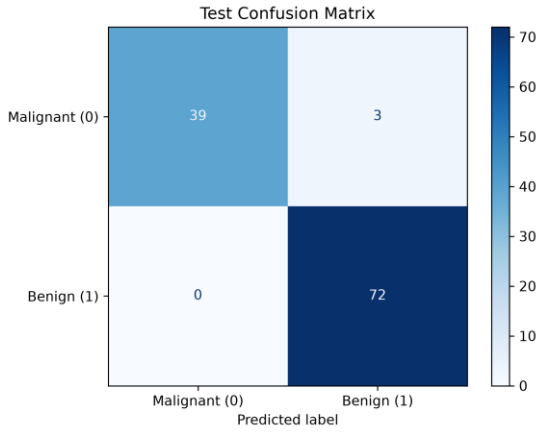


Figure 11 Breast Cancer test confusion matrix

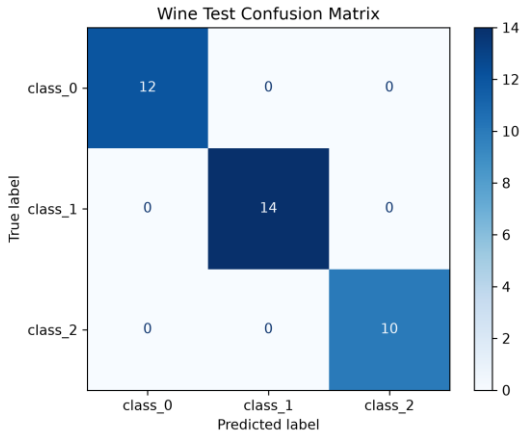


Figure 12 Wine test confusion matrix

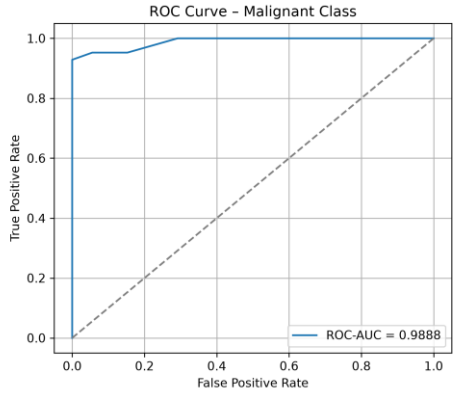


Figure 13 Breast Cancer malignant-class ROC curve

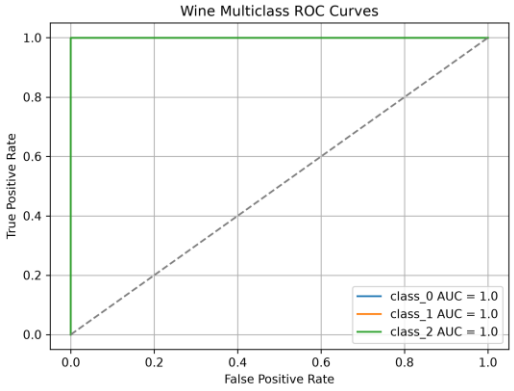


Figure 14 Wine one-versus-rest ROC curves

The Breast Cancer classifier produced three false negatives for malignant cases and no false positives. With malignant treated as positive, precision was 1.0000, recall was 0.9286 and F1 was 0.9630. The Wine classifier made no test errors, so no off-diagonal confusion was present. The error-free Wine result should be interpreted cautiously because the test subset was small.

K-Means Clustering

K-Means was fitted on training features only for both representations and all three seeds. Breast Cancer used two clusters and Wine used three clusters. True labels were used only for ARI evaluation and contingency tables, not during fitting. The two-feature figures are projections; all original features were used by the fitted models.

Dataset	Representation	Test ARI	Test Silhouette
cancer	unscaled	0.5585	0.7120
cancer	scaled	0.7354	0.3251
wine	unscaled	0.3733	0.4929
wine	scaled	0.9088	0.2711

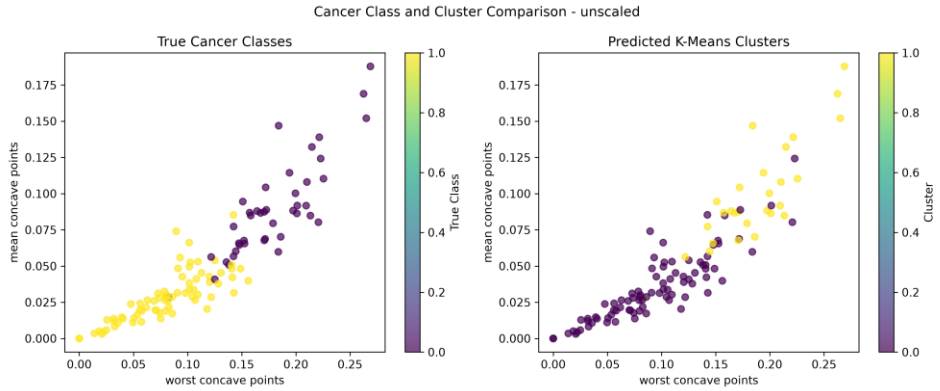


Figure 15 Breast Cancer unscaled true classes and K-Means clusters

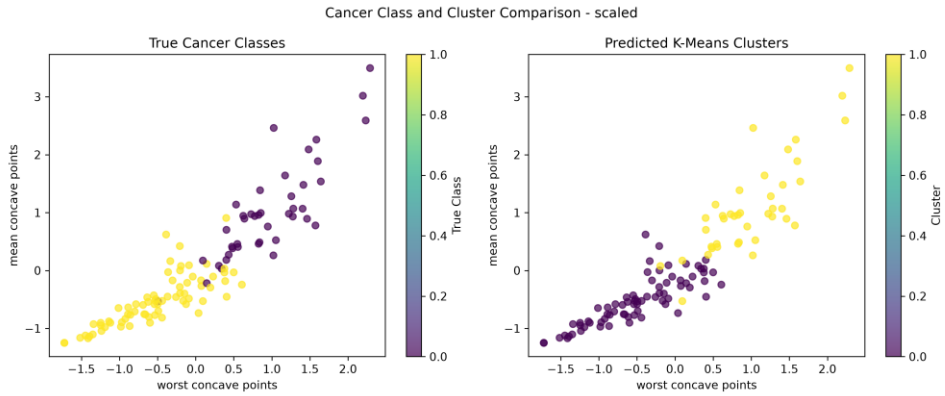


Figure 16 Breast Cancer scaled true classes and K-Means clusters

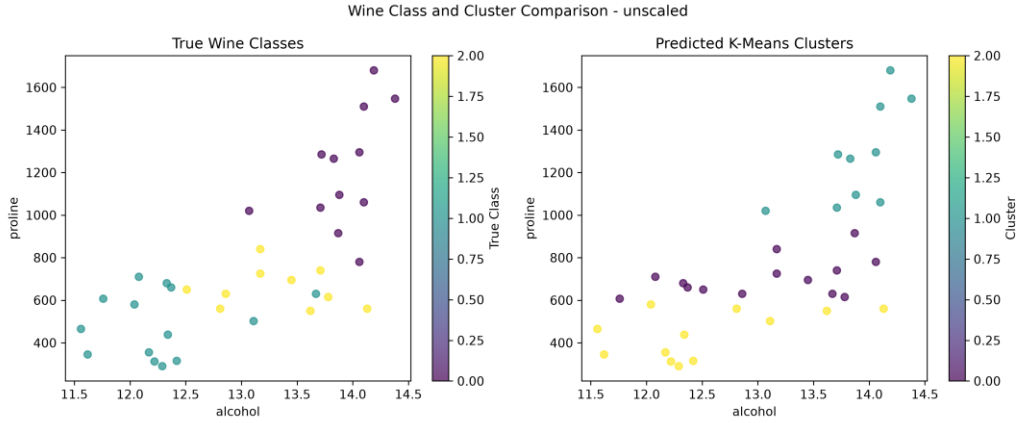


Figure 17 Wine unscaled true classes and K-Means clusters

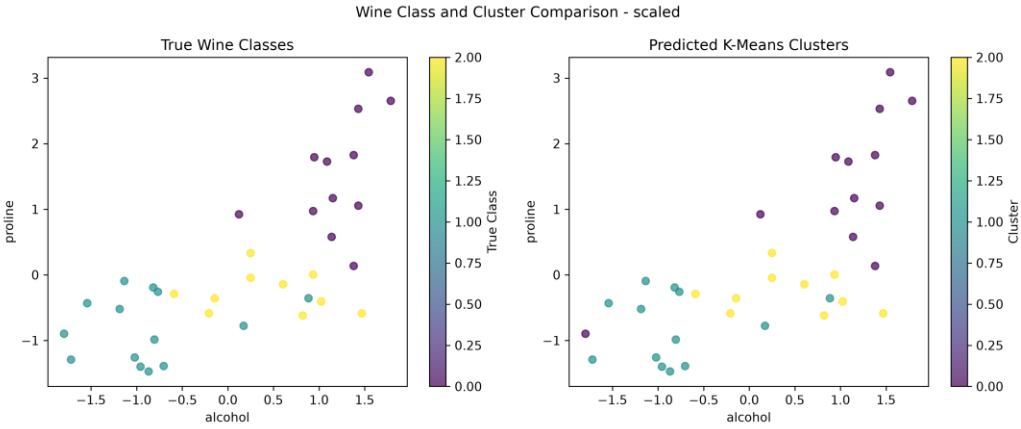


Figure 18 Wine scaled true classes and K-Means clusters

Comparing the Two Datasets

The same algorithm and representation performed best on both datasets: scaled KNN. On validation, Breast Cancer selected k=7 with mean macro-F1 0.9677 and standard deviation 0.0297; Wine selected k=11 with mean macro-F1 0.9814 and standard deviation 0.0323. On the separate final test sets, Breast Cancer achieved accuracy 0.9737 and macro-F1 0.9713, while Wine achieved accuracy and macro-F1 of 1.0000.

KNN was strongly affected by scaling because Euclidean distance is sensitive to feature magnitude. On validation, Breast Cancer KNN with k=7 improved from mean macro-F1 0.9040 unscaled to 0.9677 scaled. Wine KNN with k=11 improved from 0.6005 unscaled to 0.9814 scaled. Across classifier configurations, validation macro-F1 standard deviations ranged from 0.0013 to 0.0566, showing variation across seeds. Increasing tree depth or neighbour count did not always improve validation performance.

The main classification errors differed across datasets. Breast Cancer produced three malignant cases classified as benign, which is important because these are false negatives. Wine produced no test errors under the selected model, but this should not be interpreted as proof of perfect performance on new populations.

Scaling improved K-Means agreement with the true classes in both datasets. Breast Cancer test ARI increased from 0.5585 to 0.7354, while Wine test ARI increased from 0.3733 to 0.9088. Silhouette values decreased after scaling because distances were measured in a different feature space; therefore, silhouette values across representations should not be treated as directly equivalent evidence of model quality.

As future work, nested cross-validation or an additional untouched external test set could be investigated. This would provide a more robust estimate of generalisation while keeping model selection separate from final evaluation.

Required Run and Output Inventory

Item	Breast Cancer	Wine	Overall
Classifier fits	66	66	132
K-Means fits	6	6	12
Total model fits	72	72	144
Required figures	9	9	18

The project exports classifier_runs.csv, classifier_summary.csv, classifier_test.csv, kmeans_runs.csv, kmeans_summary.csv and kmeans_test.csv. Final test evaluations reuse pre-fitted seed-42 models and are not counted as additional fits.

Reproducibility and Submission Information

Python and package versions are recorded in the notebooks. Both notebooks should be executed from top to bottom before submission. Git hub link https://github.com/mvltay/ml-midterm Video URL h ttps://youtu.be/pCpbLbHta8M. The final project package should include both notebooks, this report, the six combined CSV files, 18 figure files, README.md, requirements.txt and AI_DISCLOSURE.md. The split summaries, scaling before-and-after checks, per-class classification reports, manual metric checks and four class-versus-cluster contingency tables are available in the notebooks rather than reproduced in this report. The raw run-level results are available in the notebooks and CSV files.